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Inventor(s)

ZHANG; Guo et al.

AEROSOL GENERATING DEVICE AND HEATING ASSEMBLY

Abstract

An aerosol generating device includes: a case with an assembling opening in one end; and a heating assembly detachably mounted in the case so as to heat an aerosol generating substrate. The heating assembly includes a heating structure. The heating structure includes a heating portion for generating infrared light waves in a power-on state, and a sleeve for the infrared light waves to pass through. The heating portion is arranged in the sleeve and is at least partially spaced apart from a tube wall of the sleeve.

Inventors: ZHANG; Guo (Shenzhen, CN), MA; Lei (Shenzhen, CN), ZHOU; Hongming (Shenzhen, CN), LI; Rihong (Shenzhen, CN), DU; Xianwu (Shenzhen, CN), CHU; Qingchen (Shenzhen, CN)

Applicant: SMOORE INTERNATIONAL HOLDINGS LIMITED (George Town, KY)

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Background/Summary

CROSS-REFERENCE TO PRIOR APPLICATION [0001] This application is a continuation of International Patent Application No. PCT/CN2023/114119, filed on Aug. 21, 2023, which claims priority to Chinese Patent Application No. 202211442116.1, filed on Nov. 17, 2022. The entire disclosure of both applications is hereby incorporated by reference herein.

FIELD

[0002] The present disclosure relates to the field of heat-not-burn atomization, and in particular, to an aerosol generating device and a heating assembly.

BACKGROUND

[0003] In the field of HNB (heat-not-burn) atomization, heating methods such as heating through a central heating element heating or a circumferential heating element are generally adopted. The usual approach is that the heating element generates heat, and then the heat is directly transferred to an aerosol generating substrate through thermal conduction. The heating assembly with the heating element is usually fixed inside the aerosol generating device and cannot be disassembled or replaced. After long-term use, it will age, be damaged, and experience surface contamination, resulting in a decrease in the taste quality of the aerosol generated by the heated aerosol generating substrate.

SUMMARY

[0004] In an embodiment, the present invention provides an aerosol generating device, comprising: a case with an assembling opening in one end; and a heating assembly detachably mounted in the case and configured to heat an aerosol generating substrate, wherein the heating assembly comprises a heating structure, wherein the heating structure comprises a heating portion configured to generate infrared light waves in a power-on state, and a sleeve for the infrared light waves to pass through, and wherein the heating portion is arranged in the sleeve and is at least partially spaced apart from a tube wall of the sleeve.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Subject matter of the present disclosure will be described in even greater detail below based on the exemplary figures. All features described and/or illustrated herein can be used alone or combined in different combinations. The features and advantages of various embodiments will become apparent by reading the following detailed description with reference to the attached drawings, which illustrate the following:

[0006] FIG. 1 is a schematic exploded structural diagram of an aerosol generating device according to a first embodiment of the present disclosure.

[0007] FIG. 2 is a cross-sectional diagram of the aerosol generating device in FIG. 1.

[0008] FIG. 3 is a schematic structural diagram of a heating assembly of the aerosol generating device in FIG. 1.

[0009] FIG. 4 is a first longitudinal cross-sectional diagram of the heating assembly in FIG. 3.

[0010] FIG. 5 is a second longitudinal cross-sectional diagram of the heating assembly in FIG. 3.

[0011] FIG. 6 is a third longitudinal cross-sectional diagram of the heating assembly in FIG. 3.

[0012] FIG. 7 is a schematic exploded structural diagram of the heating assembly in FIG. 3.

[0013] FIG. 8 is a schematic bottom structural diagram of the heating assembly in FIG. 3.

[0014] FIG. 9 is a transverse cross-sectional diagram of a heating element of the heating assembly in FIG. 3.

[0015] FIG. 10 is a transverse cross-sectional diagram of a heating element of an aerosol generating device according to a second embodiment of the present disclosure.

[0016] FIG. 11 is a transverse cross-sectional diagram of a heating element of an aerosol generating device according to a third embodiment of the present disclosure.

DETAILED DESCRIPTION

[0017] In an embodiment, the present invention provides an improved aerosol generating device and a heating assembly.

[0018] In an embodiment, the present invention provides an aerosol generating device, including a case with an assembling opening in one end, an extractor detachably mounted at the assembling opening and configured to accommodate an aerosol generating substrate, and a heating assembly detachably mounted in the case and configured to heat the aerosol generating substrate.

[0019] The heating assembly includes a heating structure; and the heating structure includes a heating portion configured to generate infrared light waves in a power-on state and a sleeve for the infrared light waves to pass through, and the heating portion is arranged in the sleeve and is at least partially spaced apart from the tube wall of the sleeve.

[0020] In some embodiments, the aerosol generating device further includes a fixing sleeve arranged in the case in a pluggable manner; and the fixing sleeve is a hollow structure with two ends running through, and the heating assembly is mounted in the fixing sleeve and is detachably arranged in the fixing sleeve.

[0021] In some embodiments, the fixing sleeve and the heating assembly are provided with a first connecting structure, and the fixing sleeve and the heating assembly are detachably connected through the first connecting structure.

[0022] In some embodiments, the first connecting structure includes a clamping buckle and a clamping hole for cooperating with the clamping buckle;

[0023] the clamping buckle is arranged on the heating assembly, and the clamping hole is provided in the side wall of the fixing sleeve and corresponds to the clamping buckle;

[0024] or, the clamping hole is provided in the heating assembly, and the clamping buckle is arranged on the inner side wall of the fixing sleeve and corresponds to the clamping hole.

[0025] In some embodiments, the fixing sleeve includes a first open end and a second open end spaced apart in the axial direction; and the first open end is configured to arrange the extractor in a partially pluggable manner, and the heating assembly is arranged close to the second open end.

[0026] In some embodiments, the first open end of the fixing sleeve is provided with an extension portion for cooperating with the assembling opening; and the extension portion is detachably connected to the case by arranging a second connecting structure.

[0027] In some embodiments, the second connecting structure includes a first magnetic member and a second magnetic member, the first magnetic member is arranged on the extension portion, and the second magnetic member is arranged in the case and corresponds to the first magnetic member.

[0028] In some embodiments, the extractor includes an accommodating cavity configured to accommodate the aerosol generating substrate, and the heating assembly is at least partially detachably inserted into the accommodating cavity.

[0029] The present disclosure further provides a heating assembly, detachably mounted in an aerosol generating device and configured to heat an aerosol generating substrate, and including a supporting base and a heating structure mounted on the supporting base, where the heating structure includes a heating portion configured to generate infrared light waves in a power-on state and a sleeve for the infrared light waves to pass through, the heating portion is arranged in the

sleeve and is at least partially spaced apart from the tube wall of the sleeve, the sleeve is provided with an opening, and the opening is provided in the supporting base.

[0030] In some embodiments, a first connecting structure detachably connected to the aerosol generating device is provided on the supporting base.

[0031] In some embodiments, the heating portion is arranged in the sleeve in a pluggable manner.

[0032] In some embodiments, the heating structure includes two conductive portions, and the two conductive portions are connected to the heating portion, led out from the opening, and detachably and electrically connected to the supporting base in a state that the heating structure is mounted on the supporting base.

[0033] In some embodiments, conductive members are arranged on the supporting base, and the conductive members correspond to the conductive portions, are detachably connected to the conductive portions, and are electrically connected to the conductive portions in the state that the heating structure is mounted on the supporting base.

[0034] In some embodiments, a separating member configured to separate and insulate the two adjacent conductive portions from each other is arranged in the supporting base.

[0035] In some embodiments, the supporting base includes a holder configured to support the heating structure;

[0036] the holder includes a first holder body and a second holder body arranged in an openable and closable manner; and the first holder body and the second holder body clamp or release the heating structure by closing or opening.

[0037] In some embodiments, the supporting base further includes a seal member, the seal member is detachably arranged between the first holder body and the second holder body, and the seal member is detachably and partially sleeved over the heating structure and configured to hermetically connect the heating structure with the first holder body and the second holder body.

[0038] In some embodiments, the seal member includes a sleeve body with two ends running through and configured to be partially sleeved over the heating structure, and a first seal portion protruding from the outer side wall of the sleeve body; and

[0039] the first seal portion is respectively clamped and fixed with the first holder body and the second holder body.

[0040] In some embodiments, the supporting base further includes a shell detachably sleeved over the holder; and

[0041] the shell is provided with a through hole for the heating structure to partially penetrate through.

[0042] In some embodiments, the seal member includes a sleeve body with two ends running through and configured to be partially sleeved over the heating structure, and a second seal portion protruding from the outer side wall of the sleeve body; and the second seal portion is located between the holder and the shell in an assembled state of the shell and the holder, and is configured to seal a gap formed between the holder and the end surface of the through hole.

[0043] In some embodiments, the shell includes a sleeve opening for mounting the holder; and

[0044] the holder includes the bottom wall, and a gap is provided between the bottom wall and the sleeve opening in an assembled state of the shell and the holder.

[0045] In some embodiments, the holder and the shell are provided with a third connecting structure.

[0046] In some embodiments, the third connecting structure includes buckle holes and clamping protrusions; the clamping protrusions protrude from the outer side wall of the holder; and the buckle holes are provided in the side wall of the shell and correspond to the clamping protrusions one to one for clamping with the clamping protrusions.

[0047] In some embodiments, the heating assembly further includes a temperature measurement structure arranged on the heating structure and detachably connected to the supporting base.

[0048] The present disclosure further provides an aerosol generating device, including the heating

assembly in the present disclosure, and a power supply assembly connected to the heating assembly.

[0049] The aerosol generating device and the heating assembly in the present disclosure have the following beneficial effects: In the aerosol generating device, by detachably arranging the heating assembly in the case, the overall replacement and cleaning of the heating assembly can be facilitated, thereby improving the taste of the aerosol generated by the heatable aerosol generating substrate.

[0050] In addition, the heating portion of the heating structure can generate infrared light waves, and the infrared light waves can pass through the sleeve to reach and heat the aerosol generating substrate. In a case that the highest working temperature of the heating portion reaches 1000° C. or above (the working temperature of traditional HNB heating elements generally does not exceed 400° C.), it will not cause the aerosol generating substrate to be overheated, thereby greatly improving the puffing taste. Moreover, the preheating time is greatly reduced, thereby greatly improving the consumer experience.

[0051] To provide a clearer understanding of the technical features, objectives, and effects of the present disclosure, the specific embodiments of the present disclosure will be described below with reference to the drawings.

[0052] FIG. 1 and FIG. 2 show an aerosol generating device according to a first embodiment of the present disclosure. The aerosol generating device **100** may adopt a low-temperature heat-not-burn method to heat an aerosol generating substrate, and has the advantages of good atomization stability and excellent puffing taste. In some embodiments, the aerosol generating substrate may be arranged in the aerosol generating device **100** in a pluggable manner. The aerosol generating substrate may be cylindrical. Specifically, the aerosol generating substrate may be a filamentous or sheet-like solid material made from leaves and/or stems of plants, and aroma components may be further added to the solid material.

[0053] Referring to FIG. 1 and FIG. 2, further, in this embodiment, the aerosol generating device **100** includes a heating assembly **10** and a power supply assembly **20**. The heating assembly **10** may be partially inserted into the aerosol generating substrate. Specifically, it may be partially inserted into a substrate section of the aerosol generating substrate, and generates thermal radiation to heat the substrate section of the aerosol generating substrate in a power-on state, causing it to be atomized to generate an aerosol. In this embodiment, the thermal radiation may be thermal infrared radiation. The heating assembly **10** has the advantages of easiness in assembling, simple structure, high atomization efficiency, strong stability, and long service life. The power supply assembly **20** is configured to supply power to the heating assembly **10**.

[0054] In this embodiment, the heating assembly **10** includes a heating structure **11** and a supporting base **12**. The heating structure **11** is mounted on the supporting base **12**. In this embodiment, the heating structure **11** and the supporting base **12** are detachably mounted to facilitate the replacement and maintenance of the heating structure **11**. The supporting base **12** may be mechanically and electrically connected to the heating structure **11**. It can not only support the heating structure **11**, but also be electrically connected to the heating structure **11** when the heating structure **11** is mounted on it, thereby electrically connecting the heating structure **11** to the power supply assembly **20**. As can be understood, in some other embodiments, the supporting base **12** may only play a supporting role.

[0055] Referring to FIG. 3 to FIG. 6, in this embodiment, the heating structure **11** includes a sleeve **111** and a heating element **112**. The sleeve **111** covers at least part of the heating element **112** and may allow light waves to pass through to reach the aerosol generating substrate. Specifically, in this embodiment, the sleeve **111** may allow infrared light waves to pass through, thereby facilitating the heating element **112** to radiate heat to heat the aerosol generating substrate. Specifically, in this embodiment, a gap is provided between the inner wall of the sleeve **111** and the heating element **112**. In a power-on state, the heating element quickly heats to increase the temperature to about

1000° C. within 1-3 s, the surface temperature of the sleeve **111** can be controlled below 350° C., and the overall atomization temperature of the aerosol generating substrate is controlled at 300-350° C., thereby achieving the precise atomization of the aerosol generating substrate in a wave band of 2-5 μm . In the present disclosure, the highest working temperature of the heating element ranges from 500° C. to 1300° C., which is much higher than the highest working temperature of the heating element in the existing technology.

[0056] In this embodiment, the sleeve **111** may be a quartz glass tube. Of course, as can be understood, in some other embodiments, the sleeve **111** is not limited to a quartz tube and may be any other window material that allows light waves to pass through, such as infrared transparent glass, transparent ceramics, or diamond.

[0057] Referring to FIG. 7 to FIG. 9 together, in this embodiment, the sleeve **111** is a hollow tubular body. Specifically, the sleeve **111** includes a tubular body **1111** with a circular cross section, and a pointed structure **1112** arranged at one end of the tubular body **1111**. Of course, as can be understood, in some other embodiments, the cross section of the tubular body **111** is not limited to be circular. The tubular body **1111** is a hollow structure with an opening **1110** in one end. The sleeve **111** may be mounted on the supporting base **12**. Specifically, the sleeve **111** may be partially inserted into the supporting base **12**. Its opening may be located in the supporting base **12**. The pointed structure **1112** is arranged at the end of the tubular body **1111** away from the opening **1110**. By arranging the pointed structure **1112**, it is convenient for at least part of the heating structure **111** to be plugged into the aerosol generating substrate. In this embodiment, an accommodating cavity **1113** is formed on the inner side of the sleeve **111**. The accommodating cavity **1113** is a cylindrical cavity and may be non-sealed. When the heating element **112** is mounted in it, the accommodating cavity **1113** does not need to be vacuumed or filled with inert gas. In this embodiment, the sleeve **111** further includes a positioning portion **1114**. The positioning portion **1114** is arranged at the opening **1110** of the tubular body **111** and may extend radially outwards along the tubular body **111** to form a positioning flange for the mounting and positioning of the sleeve **111** and the supporting base **12**. In this embodiment, the positioning portion **1114** may be integrally formed with the tubular body **111**. Of course, as can be understood, in some other embodiments, the positioning portion **1114** may be detachably assembled with the sleeve **111** by sleeving, screwing, or clamping, for example. In this embodiment, an air gap is provided between the inner wall of the sleeve **111** and the heating element **112**. The air gap may be filled with air. By providing the air gap, there can be no direct contact between the sleeve **111** and the heating element **112**.

[0058] In this embodiment, the number of the heating element **112** may be one, which may be longitudinally arranged and wound to form a heating portion **1120** which is spiral on the whole. Specifically, the heating element **112** may be cylindrical on the whole and may be wound to form a single-spiral structure, double-spiral structure, M-shaped structure, N-shaped structure, or structure with any other shape. Of course, as can be understood, in some other embodiments, the number of the heating element **112** is not limited to one, and may be two or more. The shape of the heating element **112** is not limited to a cylindrical shape. In some embodiments, the shape of the heating element **112** may be a sheet shape.

[0059] In this embodiment, the heating portion **1120** may be placed in the sleeve **111**, spaced apart from the tube wall of the sleeve **111** on the whole, and configured to generate infrared light waves in a power-on state. The infrared light waves may pass through the sleeve **111** to reach the aerosol generating substrate. Of course, as can be understood, in some other embodiments, the heating portion **1120** may also be partially spaced apart from the tube wall of the sleeve **111**. In this embodiment, the heating portion **1120** may be in a longitudinally elongated spiral shape. Of course, as can be understood, in some other embodiments, the heating portion **1120** is not limited to a spiral shape.

[0060] In this embodiment, one end of the heating portion **1120** is provided with a conductive portion **1121**. The conductive portion **1121** is connected to the heating portion **1120**, and may be led

out from the opening **1110** of the sleeve **111**, penetrate through a base, and be electrically connected to the power supply assembly **20**. In this embodiment, the conductive portion **1121** may be fixed to the heating portion **1120** through welding to form an integrated structure. Of course, as can be understood, in some other embodiments, the heating portion **1120** may be integrally formed with the conductive portion **1121**. In this embodiment, the number of the conductive portions **1121** may be two. The two conductive portions **1121** may be spaced apart, be respectively connected to two ends of the heating portion **1120**, extend towards the same end, and penetrate through the sleeve **111** from the opening **1110** in one end of the sleeve **111**. In this embodiment, the conductive portion **1121** may be a lead wire, which may be welded to the heating portion **1120**. Of course, as can be understood, in some other embodiments, the conductive portion **1121** is not limited to a lead wire and may be any other conductive structure. By arranging the conductive portion **1121** at one end of the heating portion **1120** and leading it out from the sleeve **111**, the entire heating structure **11** can be conveniently assembled, thereby simplifying the assembling process. During assembling, the heating structure **11** may be mounted on the supporting base **12**, and then enabled to be in contact with a conductive member **124** located in the supporting base **12**.

[0061] In this embodiment, the heating element **112** forming the heating portion **1120** includes a heating layer **1122** and a thermal radiation layer **1124**. The heating layer **1122** may generate heat in a power-on state. The thermal radiation layer **1124** is arranged on the outer surface of the heating layer **1122** and is configured to radiate the heat generated by the heating layer **1122**. In this embodiment, the heating layer **1122** and the thermal radiation layer **1124** are concentrically distributed on the cross section of the heating portion **1120**.

[0062] In this embodiment, the heating layer **1122** may be cylindrical on the whole. Specifically, the heating layer **1122** may be a heating wire. Of course, as can be understood, in some other embodiments, the heating layer **1122** may not be limited to be cylindrical, but may be in a sheet shape, that is, the heating layer **1122** may be a heating sheet. The heating layer **1122** includes a metal substrate with high-temperature oxidation resistance. The metal substrate may be a metal wire. Specifically, the heating layer **1122** may be a nickel-chromium alloy substrate (such as nickel-chromium alloy wire), iron-chromium-aluminum alloy substrate (such as iron-chromium-aluminum alloy wire) or the like made of a metal material with good high-temperature oxidation resistance, high stability, and good deformation resistance. In this embodiment, the radial size of the heating layer **1122** may be 0.15 mm-0.8 mm.

[0063] In this embodiment, the heating element **112** further includes an antioxidant layer **1123**, and the antioxidant layer **1123** is formed between the heating layer **1122** and the thermal radiation layer **1124**. Specifically, the antioxidant layer **1123** may be an oxide film, the heating layer **1122** undergoes high-temperature heat treatment to form a dense oxide film on its own surface, and the oxide film forms the antioxidant layer **1123**. Of course, as can be understood, in some other embodiments, the antioxidant layer **1123** is not limited to include an oxide film formed by itself. In some other embodiments, it may be an antioxidant coating layer coated on the outer surface of the heating layer **1122**. By forming the antioxidant layer **1123**, it can ensure that the heating layer **1122** is not or is rarely oxidized when heated in the air environment, thereby improving the stability of the heating layer **1122**. Therefore, there is no need to vacuum, fill inert gas or reducing gas into the first accommodating cavity **1113**, nor to seal the opening **1110**, thereby simplifying the assembling process of the entire heating structure **11** and reducing the manufacturing cost. In this embodiment, the thickness of the antioxidant layer **1123** may be selectively 1 μm -150 μm . When the thickness of the antioxidant layer **1123** is less than 1 μm , the heating layer **1122** is easily oxidized. When the thickness of the antioxidant layer **1123** is greater than 150 μm , it will influence the heat conduction between the heating layer **1122** and the thermal radiation layer **1124**.

[0064] In this embodiment, the thermal radiation layer **1124** may be an infrared layer. The infrared layer may be an infrared layer forming substrate formed on the side of the antioxidant layer **1123** away from the heating layer **1122** under high-temperature heat treatment. In this embodiment, the

infrared layer forming substrate may be a silicon carbide, spinel, or composite substrate. Of course, as can be understood, in some other embodiments, the thermal radiation layer **1124** is not limited to an infrared layer. In some other embodiments, the thermal radiation layer **1124** may be a composite infrared layer. In this embodiment, the infrared layer may be formed on the side of the antioxidant layer **1123** away from the heating layer **1122** through dip coating, spray coating, brush coating, and other methods. The thickness of the thermal radiation layer **1124** may be 10 μm -300 μm . When the thickness of the thermal radiation layer **1124** is 10 μm -300 μm , the thermal radiation effect is better, and the atomization efficiency and taste of the aerosol generating substrate are better. Of course, as can be understood, in some other embodiments, the thickness of the thermal radiation layer **1124** is not limited to 10 μm -300 μm .

[0065] In this embodiment, the heating assembly **11** further includes an insulating member **113**, the insulating member **113** is cylindrical, and its radial size may be smaller than the radial size of the accommodating cavity **1113**. The insulating member **113** may fully or partially penetrate into the accommodating cavity **1113** through the opening **1110** of the sleeve **111**, thereby separating the two conductive portions **1121**, that is, insulating the two conductive portions **1121**. In this embodiment, the insulating member **113** is provided with through holes **1131**, the number of the through holes **1131** is two, the two through holes **1131** correspond to the two conductive portions **1121** one to one, and the through holes **1131** may extend along the axial direction of the insulating member **113** and be configured to allow the conductive portions **1121** to penetrate through for electrical connection with the supporting base **12**. In some embodiments, the insulating member **113** may not be limited to be cylindrical. In some embodiments, the insulating member **113** may be an insulating partition, and the through holes **1131** may be omitted. In some embodiments, the insulating member **113** may be a ceramic body, quartz tube, or any other insulating structure.

[0066] In this embodiment, the supporting base **12** may support the sleeve **111** and the heating portion **1120**, and may be detachably connected to the aerosol generating device by arranging a first connecting structure. The supporting base **12** includes a holder **121**, a shell **122**, and a seal member **123**. The holder **121** is configured to support the heating structure **11**. The shell **122** may be sleeved over the outer periphery of the holder **121**. The seal member **123** may be mounted on the holder **121** and configured to hermetically connect the heating structure **11** with the holder **121** and the shell **122**.

[0067] In this embodiment, the holder **121** includes a first holder body **121a** and a second holder body **121b** that are openable and closable. By arranging the first holder body **121a** and the second holder body **121b** that are openable and closable, the mounting and dismounting of the heating structure **11** can be facilitated. In some embodiments, splicing the first holder body **121a** and the second holder body **121b** can form a rectangular solid structure. Of course, as can be understood, in some other embodiments, it is not limited to form a rectangular solid shape by splicing the first holder body **121a** and the second holder body **121b**. In some other embodiments, splicing the first holder body **121a** and the second holder body **121b** may form a cylindrical or other shape.

[0068] In this embodiment, one end of each of the first holder body **121a** and the second holder body **121b** is provided with an end plate **1210**, separating plates **1212** are correspondingly arranged in both the first holder body **121a** and the second holder body **121b**, the separating plates **1212** divide the holder **121** into upper and lower spaces, a clamping groove **1211** for cooperating with the seal member **123** is formed in the space close to the end plate **1210**, each separating plate **1212** is provided with a semi-cylindrical first avoidance hole **1216**, and the two separating plates **1212** of the first holder body **121a** and the second holder body **121b** are arranged opposite to each other, and the first avoidance holes **1216** are spliced to form a first through hole for the heating structure **11** to pass through.

[0069] In this embodiment, the holder **121** further includes the bottom wall **1213**, and the bottom wall **1213** is arranged on the first holder body **121a**. Of course, as can be understood, in some other embodiments, the bottom wall **1213** is not limited to be arranged on the first holder body **121a**, but

may also be arranged on the second holder body **121b**.

[0070] In this embodiment, a separating member **1215** is arranged in the supporting base **12**. Specifically, the separating member **1215** protrudes from the bottom wall **1213** and is integrally formed with the bottom wall **1213**. It may be a rib plate and be configured to separate the two adjacent conductive portions **1121** and insulate the two conductive portions **1121**.

[0071] In this embodiment, the first holder body **121a** and the second holder body **121b** are provided with limiting plate **1216** for limiting the heating structure **11**, the limiting plates **1216** are arranged below the separating plates **1212** and are spaced apart from the separating plates **1212**, and each limiting plate **1216** is provided with a semi-cylindrical second avoidance hole **1217**. When the first holder body **121a** and the second holder body **121b** are spliced, the two second avoidance holes **1217** in the two limiting plates **1216** are spliced to form a second through hole for the heating structure **11** to pass through. The radial size of the second through hole is smaller than the radial size of the positioning portion **1114** at one end of the sleeve **111**, thereby cooperating with the positioning portion **1114** to position the heating structure **11**.

[0072] In this embodiment, the shell **122** may be sleeved over the outer periphery of the holder **121** after the heating structure **11** is assembled with the holder **121**, thereby playing a role of fixing the first holder body **121a** and the second holder body **121b**, so that the heating structure **11** and the supporting base **12** form an integrated structure. In this embodiment, the shape and size of the shell **122** may be adapted to the holder **121**. In this embodiment, the shell **122** is roughly in a rectangular solid shape and is a hollow structure with a sleeve opening **1221** in one end. A gap is provided between the sleeve opening **1221** and the bottom wall **1213**, thereby preventing the residual aerosol in the case **21** from condensing and forming condensate that affects the normal operation of the heating structure **11**.

[0073] In this embodiment, the shell **122** may be detachably connected with the holder **121**. Specifically, in this embodiment, the shell **122** and the holder **121** are provided with a third connecting structure **125** and the shell **122** and the holder **121** are detachably connected through the third connecting structure **125**. In this embodiment, the third connecting structure **125** includes buckle holes **1222** and clamping protrusions **1214**. The clamping protrusions **1214** protrude from the outer side wall of the holder **121**. Specifically, the number of the clamping protrusions **1214** is two. The two clamping protrusions **1214** are correspondingly arranged on the outer side walls of the first holder body **121a** and the second holder body **121b**. The buckle holes **1222** are provided in the side wall of the shell **122** and the number is two. The two buckle holes **1222** correspond to the two clamping protrusions **1214** one to one. When the shell **122** is assembled with the holder **121**, the clamping protrusions **1214** can be clamped into the buckle holes **1222**, thereby connecting and fixing the shell **122** and the holder **121**.

[0074] In this embodiment, the side of the shell **122** opposite to the sleeve opening **1221** is provided with a blocking wall **1223**, and the shell **122** is provided with a through hole **1224**. Specifically, the through hole **1224** is provided in the blocking wall **1223** and can allow the heating structure **11** to partially penetrate through.

[0075] In this embodiment, the seal member **123** is detachably arranged between the first holder body **123a** and the second holder body **123b**, and the seal member **123** is detachably sleeved over the heating structure **11**. Specifically, it may be sleeved over the outer periphery of a partial section of the sleeve **111**, and configured to hermetically connect the heating structure **11** with the first holder body **121a** and the second holder body **121b**. In this embodiment, the seal member **123** may be a silica gel member, which serves to prevent vibration and prevent damage when the sleeve **111** and the holder **121** are assembled. Of course, as can be understood, in some other embodiments, the seal member **123** is not limited to a silica gel member.

[0076] In this embodiment, the seal member **123** is a hollow structure with two ends running through, a channel **1230** may be formed on the inner side, and the channel **1230** may allow the sleeve **11** to pass through. In this embodiment, the seal member **123** includes a sleeve body **1231**, a

first seal portion **1232**, and a second seal portion **1233**. The sleeve body **1231** is cylindrical, is a hollow structure with two ends running through, and is configured to be sleeved over part of the heating structure **11**. The first seal portion **1232** and the second seal portion **1233** protrude from the outer side wall of the sleeve body **1231**, and are spaced apart along the axial direction of the sleeve body **1231**. The first seal portion **1232** may be arranged along the circumferential direction of the sleeve body **1232**, and may be roughly in a circular ring shape. The first seal portion **1232** is respectively clamped and fixed with the first holder body **121a** and the second holder body **121b**. Specifically, the first seal portion **1232** may be respectively clamped into the clamping grooves **1211** in the first holder body **121a** and the second holder body **121b**. The second seal portion **1233** protrudes from the outer side wall of the sleeve body **1231** and is roughly in a circular ring shape, and its radial size is larger than the radial size of the first seal portion **1232**. The second seal portion **1233** may be placed on the side where the end wall **1210** of the first holder body **121a** and the second holder body **121b** is opposite to the clamping groove **1211**. In an assembled state of the shell **12** and the holder **121**, the second seal portion **1233** is located between the shell **122** and the holder **121**. Specifically, it is located between the blocking wall **1223** and the end wall **1210**, and is configured to seal a gap formed between the holder **121** and the end surface of the through hole **1224**. In this embodiment, the sleeve body **1231**, the first seal portion **1232**, and the second seal portion **1233** are an integrally formed structure, forming multiple sealing structures. That is, by arranging the seal member **123**, the sealing among the shell **122**, the heating structure **11**, and the holder **121** can be achieved, thereby simplifying the sealing process, reducing the manufacturing cost, and preventing condensate from flowing into the holder **121**.

[0077] In this embodiment, multiple conductive members **124** are arranged on the supporting base **12**. Specifically, the conductive members **124** are arranged corresponding to the conductive portions **1121** one to one. Of course, as can be understood, in some other embodiments, the number of the conductive member **124** may be one. The conductive member **124** may be an electrode pillar. The multiple conductive members **124** are spaced apart on the bottom wall **1213**, and may be detachably connected to the conductive portions **1121**. Specifically, in a state that the heating structure **11** is mounted on the supporting base **12**, the conductive portion **1121** may be wound around the conductive member **124**, thereby being electrically connected to the conductive member **124**. In this embodiment, the conductive member **124** may be electrically connected with a power supply in the power supply assembly **20** through contact, thereby electrically connecting the heating structure **11** with the power supply assembly **20** and facilitating the replacement of the heating element **122** when the heating element **112** reaches its service life. In this embodiment, the number of the conductive members **124** is two groups, one group is electrically connected to the heating structure **11**, and the other group may be connected to a temperature measurement structure **13**. Of course, as can be understood, in some other embodiments, the number of the conductive member **124** may be one group, and the heating structure **11** and the temperature measurement structure **13** may share one group of conductive members **124**.

[0078] In this embodiment, the heating assembly **10** further includes a temperature measurement structure **13**. The temperature measurement structure **13** is arranged on the heating structure **11** and may be detachably connected to the supporting base **12**. In this embodiment, the temperature measurement structure **13** may be sleeved over the outer periphery of a partial section of the sleeve **111**, and may be detachably connected to the conductive member **124** in the supporting base **12** and electrically connected to it when connected thereto. In this embodiment, the temperature measurement structure **13** is sleeved over a position on the sleeve **111** corresponding to a position of connection between the heating portion **1120** and the conductive portions **1121**, and includes a temperature measurement film **131** and lead wires **132**. The temperature measurement film **131** may be sleeved over the outer side wall of the sleeve **111**. The number of the lead wire **132** is two. The two lead wires **132** are spaced apart, one end of each lead wire is connected to the temperature measurement film **131**, the other end of each lead wire may be connected to the conductive

member **132** in the supporting base **12**, and they may be wound around the corresponding conductive members **132** for electrical connection and signal transmission. In some embodiments, the lead wires **132** may be connected to the temperature measurement film **131** by welding or crimping.

[0079] When the heating assembly **10** is assembled, the temperature measurement structure **13** may be first sleeved over the outer periphery of the sleeve **111**, and then the seal member **123** is sleeved over the sleeve **111** of the heating structure **11**. The first holder body **121a** is clamped at the first seal portion **1232** of the seal member **123**. Then, the conductive portions **1121** of the heating structure **11** and the lead wires **132** of the temperature measurement structure **13** are wound around the corresponding conductive members **124**. Then, the second holder body **121b** is clamped at the second seal portion **1232**. Finally, the overall structure formed by the holder **121** and the heating structure **11** is inserted into the sleeve opening **1221** in the shell **122**, so that the second seal portion **1233** is pressed against the blocking wall **1223** of the shell **122** and the end wall **1210** of the holder **121**, the seal member **123** and the heating structure **11** are enabled to partially penetrate through the through hole **1224**, and the clamping protrusions **1214** on the outer side of the holder **121** are clamped into the buckle holes **1221** in the shell **122**. If the heating structure **11** needs to be disassembled, the shell **122** is pushed out towards the direction of the pointed structure **1112** of the heating structure **11**, then the first holder body **121a** and the second holder body **121b** are separated from the seal member **123**, and finally the conductive portions **1121** and the lead wires **132** of the temperature measurement structure **13** are disconnected from the conductive members **124**.

[0080] In some embodiments, the power supply assembly **20** includes a case **21**, a holder **22**, and a power supply **23**; and the case **21** may be cylindrical, with a hollow structure inside and an assembling opening **211** in one end. The holder **22** is accommodated in the case **21**, is configured to mount the power supply **23** and can play a supporting role. The holder **22** is provided with a mounting cavity **221**. The mounting cavity **221** may be configured to mount the heating assembly **10**. The mounting cavity **221** is communicated with the assembling opening **211** and is provided with a communicating opening **2210** communicated with the assembling opening **211**. The power supply **23** is mounted on the holder **22**, may be mechanically and/or electrically connected to the heating assembly **10**, and is configured to supply power to the heating assembly **10**.

[0081] In some embodiments, the aerosol generating device further includes an extractor **30**, and the extractor **30** may be configured to accommodate the aerosol generating substrate. The extractor **30** may be detachably mounted at the assembling opening **211**. In some embodiments, the extractor **30** includes a top cover **31** detachably covering the assembling opening **211**, and an accommodating tube **32** with one end connected to the top cover **31**. The accommodating tube **32** may be integrally formed with the top cover **31**. The accommodating tube **32** may be a structure with two ends running through. An accommodating cavity **320** is formed on the inner side and configured to accommodate the aerosol generating substrate. The heating assembly is at least partially and detachably inserted into the accommodating cavity **320**, and may be inserted into the aerosol generating substrate to heat the aerosol generating substrate. The accommodating tube **32** of the extractor **30** is arranged in the mounting cavity **221** in a pluggable manner, and may be pulled out of the case **21** as a whole.

[0082] In some embodiments, the aerosol generating device further includes a fixing sleeve **40**. The fixing sleeve **40** is arranged in the case **21** in a pluggable manner, may be detachably assembled with the extractor **30**, and may form a covering assembly together with the extractor **30**.

Specifically, the fixing sleeve **40** may be arranged the mounting cavity **221** in a pluggable manner, and may be configured to mount the heating assembly **10**. Specifically, the heating assembly **10** is mounted in the fixing sleeve **40** and is detachably arranged in the fixing sleeve **40**. In this embodiment, the fixing sleeve **40** is a hollow structure with two ends running through, and includes a first open end **41** and a second open end **42** spaced apart in the axial direction; and a plugging channel configured to allow the extractor to be plugged is formed on the inner side of the fixing

sleeve **40**. The first open end **41** may be configured to allow the accommodating tube of the extractor **30** to be plugged into the fixing sleeve **40**. The heating assembly **10** may be arranged close to the second open end **42** and may be withdrawn from the second open end **42**.

[0083] In this embodiment, the fixing sleeve **40** and the heating assembly **10** may be detachably connected by arranging a first connecting structure **50**. The first connecting structure **50** may include a clamping buckle **1226** and a clamping hole **43**. The clamping buckle **1226** may be arranged on the heating assembly **10**. Specifically, in this embodiment, the clamping buckle **1226** protrudes from the outer side wall of the shell **12**, and the clamping hole **43** is provided in the side wall of the fixing sleeve **40** and corresponds to the clamping buckle **1226**. When the heating assembly **10** needs to be mounted into the aerosol generating device, the heating assembly **10** is inserted into the fixing sleeve **40** from the second open end **42** of the fixing sleeve **40**, the clamping buckle **1226** is clamped into the clamping hole **43**, then it is inserted into the accommodating tube **31** of the extractor **30**, and finally the aerosol generating substrate is inserted. When the heating assembly **10** needs to be disassembled or replaced, the extractor **30** and the entire fixing sleeve **40** are pulled out of the mounting cavity **221**, then the extractor **30** and the fixing sleeve **40** are separated, then the clamping buckle **1226** is withdrawn from the clamping hole **43**, and finally the heating assembly **10** is withdrawn from the second open end **42** of the fixing sleeve **40**. In some other embodiments, the clamping buckle **1226** may protrude from the inner wall of the fixing sleeve **40**; and the clamping hole **43** may be provided in the heating assembly **10**, is specifically located in the side wall of the shell **12** of the heating assembly **10**, and corresponds to the clamping buckle **1226** for clamping with the clamping buckle **1226**. Of course, as can be understood, in some other embodiments, the first connecting structure **50** may not be limited to a clamping structure. In some other embodiments, the first connecting structure **50** may be a magnetic attraction structure, a threaded structure, a guide slider and sliding groove fitting structure, or any other structure.

[0084] In this embodiment, the first open end **41** of the fixing sleeve **40** is provided with an extension portion **44**, the extension portion **44** may extend radially outwards along the first open end **41**, and its shape and size may be adapted to the shape and size of the assembling opening. The extension portion **44** may cover the assembling opening. The extension portion **44** may also be detachably connected to the top cover **31** of the extractor **30** by clamping or magnetic attraction.

[0085] In this embodiment, the fixing sleeve **40** and the case **21** may be detachably connected by arranging a second connecting structure **60**. In this embodiment, the second connecting structure **60** includes a first magnetic member **61** and a second magnetic member, the first magnetic member **61** is arranged on the extension portion **44**, and the second magnetic member is arranged in the case **21**, corresponds the first magnetic member **61**, is mounted on the holder **22**, and may be integrally formed on a top wall of the holder **22**. Of course, as can be understood, in some other embodiments, the second magnetic member may also be arranged independently of the holder **22**. When the fixing sleeve **40** is inserted into the mounting cavity **221** of the holder **22**, the fixing sleeve **40** may be fixed at the assembling opening **211** of the case **21** through the first magnetic member **61** and the second magnetic member. As can be understood, in some other embodiments, the second connecting structure **60** is not limited to a magnetic structure, and may be a clamping structure, a threaded structure or any other structure.

[0086] FIG. **10** shows an aerosol generating device according to a second embodiment of the present disclosure, the difference between the second embodiment and the first embodiment lies in that the thermal radiation layer **1124** is a composite infrared layer. The composite infrared layer may be formed by compounding an infrared layer forming substrate with a bond configured to bond with the antioxidant layer **1123**. Specifically, the bond may be glass powder. The composite infrared layer may be a glass powder composite infrared layer. The reason for adopting glass powder is that glass powder can melt at high temperatures to bond the antioxidant layer **1123** with the infrared layer forming substrate and block the gap in the infrared layer forming substrate, thereby further improving the breakdown resistance. The glass powder composite infrared layer

may be prepared by adding glass powder to the infrared layer forming substrate (such as silicon carbide or spinel), performing compounding, then coating the mixture onto the side of the antioxidant layer **1123** away from the heating layer **1122** through dip coating, spray coating or brush coating, performing heat treatment in a tunnel furnace, then placing the obtain material in a heating furnace, increasing the temperature to 1000-1200° C. according to certain temperature increase speed, preserving the heat, and then cooling to room temperature with the furnace.

[0087] FIG. **11** shows an aerosol generating device according to a third embodiment of the present disclosure, the difference between the third embodiment and the first embodiment lies in that the heating element **112** further includes a bonding layer **1125** arranged between the antioxidant layer **1123** and the thermal radiation layer **1124**. The bonding layer **1125** may be configured to prevent the heating layer **1122** from being broken down locally, thereby further improving the bonding strength between the antioxidant layer **1123** and the thermal radiation layer **1124**. In some embodiments, the bond in the bonding layer **1125** may be glass powder, that is, the bonding layer **1125** may be a glass powder layer.

[0088] In some embodiments, the bond may also be added to the thermal radiation layer **1124**, and the melting point of the glass powder selectable for the bonding layer **1125** is higher than the melting point of the glass powder in the thermal radiation layer **1124**.

[0089] While the invention has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. It will be understood that changes and modifications may be made by those of ordinary skill within the scope of the following claims. In particular, the present invention covers further embodiments with any combination of features from different embodiments described above and below. Additionally, statements made herein characterizing the invention refer to an embodiment of the invention and not necessarily all embodiments.

[0090] The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

Claims

1. An aerosol generating device, comprising: a case with an assembling opening in one end; and a heating assembly detachably mounted in the case and configured to heat an aerosol generating substrate, wherein the heating assembly comprises a heating structure, wherein the heating structure comprises a heating portion configured to generate infrared light waves in a power-on state, and a sleeve for the infrared light waves to pass through, and wherein the heating portion is arranged in the sleeve and is at least partially spaced apart from a tube wall of the sleeve.
2. The aerosol generating device of claim 1, further comprising: a fixing sleeve arranged in the case in a pluggable manner, wherein the fixing sleeve comprises a hollow structure with two ends running through, and wherein the heating assembly is mounted in the fixing sleeve and is detachably arranged in the fixing sleeve.
3. The aerosol generating device of claim 2, wherein the fixing sleeve and the heating assembly are

provided with a first connecting structure, and wherein the fixing sleeve and the heating assembly are detachably connected through the first connecting structure.

4. The aerosol generating device of claim 3, wherein the first connecting structure comprises a clamping buckle and a clamping hole configured to cooperate with the clamping buckle, the clamping buckle is arranged on the heating assembly, and the clamping hole is provided in a side wall of the fixing sleeve and corresponds to the clamping buckle, or, the clamping hole is provided in the heating assembly, and the clamping buckle is arranged on an inner side wall of the fixing sleeve and corresponds to the clamping hole.

5. The aerosol generating device of claim 2, further comprising: an extractor detachably mounted at the assembling opening and configured to accommodate the aerosol generating substrate, wherein the fixing sleeve comprises a first open end and a second open end spaced apart in an axial direction, wherein the first open end is configured to arrange the extractor in a partially pluggable manner, and wherein the heating assembly is arranged close to the second open end.

6. The aerosol generating device of claim 5, wherein the first open end of the fixing sleeve is provided with an extension portion configured to cooperate with the assembling opening, and wherein the extension portion is detachably connected to the case by arranging a second connecting structure.

7. The aerosol generating device of claim 6, wherein the second connecting structure comprises a first magnetic member and a second magnetic member, wherein the first magnetic member is arranged on the extension portion, and wherein the second magnetic member is arranged in the case and corresponds to the first magnetic member.

8. The aerosol generating device of claim 5, wherein the extractor comprises an accommodating cavity configured to accommodate the aerosol generating substrate, and wherein the heating assembly is at least partially detachably inserted into the accommodating cavity.

9. A heating assembly detachably mounted in an aerosol generating device and configured to heat an aerosol generating substrate, the heating assembly comprising: a supporting base; and a heating structure mounted on the supporting base, wherein the heating structure comprises a heating portion configured to generate infrared light waves in a power-on state and a sleeve for the infrared light waves to pass through, wherein the heating portion is arranged in the sleeve and is at least partially spaced apart from a tube wall of the sleeve, wherein the sleeve is provided with an opening, and wherein the opening is provided in the supporting base.

10. The heating assembly of claim 9, wherein a clamping buckle detachably connected to the aerosol generating device is provided on the supporting base.

11. The heating assembly of claim 9, wherein the heating portion is arranged in the sleeve in a pluggable manner.

12. The heating assembly of claim 9, wherein the heating structure comprises two conductive portions, and wherein the two conductive portions are connected to the heating portion, led out from the opening, and detachably and electrically connected to the supporting base in a state that the heating structure is mounted on the supporting base.

13. The heating assembly of claim 12, wherein conductive members are arranged on the supporting base, and wherein the conductive members correspond to the conductive portions, are detachably connected to the conductive portions, and are electrically connected to the conductive portions in the state that the heating structure is mounted on the supporting base.

14. The heating assembly of claim 12, wherein a separating member configured to separate and insulate the two adjacent conductive portions from each other is arranged in the supporting base.

15. The heating assembly of claim 9, wherein the supporting base comprises a holder configured to support the heating structure, wherein the holder comprises a first holder body and a second holder body arranged in an openable and closable manner, and wherein the first holder body and the second holder body are configured to clamp or release the heating structure by closing or opening.

16. The heating assembly of claim 15, wherein the supporting base comprises a seal member

detachably arranged between the first holder body and the second holder body, and wherein the seal member is detachably and partially sleeved over the heating structure and configured to hermetically connect the heating structure with the first holder body and the second holder body.

17. The heating assembly of claim 16, wherein the seal member comprises a sleeve body with two ends running through and is configured to be partially sleeved over the heating structure, wherein a first seal portion protrudes from an outer side wall of the sleeve body, and wherein the first seal portion is respectively clamped and fixed with the first holder body and the second holder body.

18. The heating assembly of claim 16, wherein the supporting base comprises a shell detachably sleeved over the holder, and wherein the shell is provided with a through hole for the heating structure to partially penetrate through.

19. The heating assembly of claim 18, wherein the seal member comprises a sleeve body with two ends running through, the seal member being configured to be partially sleeved over the heating structure, wherein a second seal portion protrudes from an outer side wall of the sleeve body, and wherein the second seal portion is located between the holder and the shell in an assembled state of the shell and the holder, and is configured to seal a gap formed between the holder and the end surface of the through hole.

20. The heating assembly of claim 18, wherein the shell comprises a sleeve opening configured to mount the holder, wherein the holder comprises the bottom wall, and wherein a gap is provided between the bottom wall and the sleeve opening in an assembled state of the shell and the holder.

21. The heating assembly of claim 18, wherein the holder and the shell are provided with a third connecting structure, wherein the third connecting structure comprises buckle holes and clamping protrusions, wherein the clamping protrusions protrude from the outer side wall of the holder, and wherein the buckle holes are provided in a side wall of the shell and correspond to the clamping protrusions one to one for clamping with the clamping protrusions.

22. The heating assembly of claim 9, further comprising: a temperature measurement structure arranged on the heating structure and detachably connected to the supporting base.
